James Akl

Robotics Researcher

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EDUCATION Coursework

PhD	Robotics Engineering	Worcester Polytechnic Institute	GPA 4.00/4.00	**Aug 2019**	**May 2024** (exp.)
BEng	Mechanical Engineering	Lebanese American University	GPA 3.72/4.00	**Sep 2015**	**May 2019**
Minor	Mathematics	Lebanese American University	GPA 4.00/4.00	**Sep 2015**	**May 2018**

EXPERIENCE Details

Doctoral Researcher — Worcester Polytechnic Institute
Manipulation & Environmental Robotics Laboratory — **Aug 2019 – PRESENT**

Develop strategies/algorithms for metal cutting, waste sorting, and soft robots using robot **vision**, **control**, and **machine learning**.

Teaching Assistant — Worcester Polytechnic Institute
Robotics Engineering Department — **Aug 2019 – May 2020**

Assisted **delivery**, **labs**, and **grading** of robotics courses: *Unified Robotics I: Actuation*, *Unified Robotics II: Sensing*, *Introduction to Robotics*.

Visiting Researcher — University of Ottawa
Mathematical Modeling & Physical Experiments — **Jun 2018 – Aug 2018**

Modeled the vibrations of ribbed plates, and **experimentally verified** the performance of **analytical** versus **numerical models**.

Research Assistant — Lebanese American University
Multiphysics Modeling & Numerical Simulation — **Nov 2017 – Mar 2019**

Demonstrated the viability of the *ionic buoyancy engine* for underwater propulsion using a coupled multiphysics **numerical model**.

SKILLS Projects

Robotics	Kinematics & Dynamics, Control, Motion planning, Manipulation & Grasping, Robot autonomy
Computing	Algorithms, Data structures, Computer vision, Machine learning & AI, Geometric algorithms
Software	C++, Python, Bash, Git, Linux, ROS/Gazebo, MATLAB/Simulink, Mathematica, CMake
Mathematics	Linear algebra, Probability & Statistics, Analysis & Optimization, Numerical algorithms, Proofs
Research	Surveys/reviews, Mathematical modeling, Simulations, Physical experiments, Field work
Publishing	MS Office, LaTeX, Technical writing, Photo-editing, Scientific illustration/diagrams, Video-editing
Social	Presentations, Communication (technical/non-technical audiences), Collaboration (Cross-disciplinary)
Language	English (native fluency), French (professional fluency), Arabic (professional fluency)

RESEARCH Preprints

[2022][1] **Feature-Driven Next View Planning for Robotic Cutting in Metal Scrap Recycling** *[Under Review]*

[2021][4] **ZeroWaste Dataset: Towards Deformable Object Segmentation in Extreme Clutter** *[Under Review]*

[2021][1] **Towards Robotic Metal Scrap Cutting: A Novel Workflow and Pipeline for Cutting Path Generation** ***IEEE CASE***

[2021][2] **ECNNs: Ensemble Learning Methods for Improving Planar Grasp Quality Estimation** ***IEEE ICRA***

[2020][1] **Comparing & Optimizing Analytical, Numerical & Experimental Vibration Models for [...]** ***IStructE/Elsevier Structures***

[2019][4] **Simplified Setup for the Vibration Study of Plates with Simply-supported Boundary Conditions** ***Elsevier MethodsX***

[2019][1] **Ionic Buoyancy Engines: Finite Element Modeling & Experimental Validation** ***SPIE Smart Structures***

[1–4] **Superscripts indicate author position in co-authored publications.**

AWARDS Proof

Graduated with Distinction, 2019
LAU School of Engineering

LAU Most Novel Club Award, 2018
Founded the Emerging Technologies Club

LAU Engineering Scholarship, 2015
Won a design and programming competition

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